

Introduction to SOC (Security Operations Centre)

What is a Security Operations Center (SOC)

Goal

Understand what a SOC is and its role within an organization.

Explanation

- A **Security Operations Centre (SOC)** is a centralized security team responsible for protecting the organization from cyber threats
- SOC's usually operate **24x7** because attacks can happen at any time
- The SOC continuously **monitors, detects, analyses, and coordinates responses** to security threats
- It acts as the **brain and command centre** of the cybersecurity program
- SOC visibility spans across:
 - Endpoints (laptops, servers)
 - Networks (firewalls, IDS, DNS)
 - Cloud platforms
 - User identities
 - Applications and logs
- The primary goal of a SOC is to **reduce risk, detect threats early, and limit business impact**

 In simple terms: **SOC is the team that keeps the organization safe every day**

Why Organizations Need a SOC

Goal

Understand why SOC's are critical for modern businesses.

Explanation

- Modern organizations generate **huge volumes of logs and security data**
- Attackers continuously target:
 - Customer data
 - Financial systems
 - Intellectual property
- Cyberattacks today are:
 - Automated
 - Highly sophisticated
 - Often multi-stage (initial access → lateral movement → impact)
- Manual monitoring is **not scalable or effective**
- SOC's help organizations:
 - Detect ransomware and phishing early
 - Reduce attacker dwell time
 - Meet compliance requirements (ISO 27001, SOC2, PCI-DSS)
 - Avoid downtime and reputational damage

 **Without a SOC, organizations have no real-time visibility or control**

Security Events vs Security Incidents

Goal

Clearly differentiate between security noise and real security threats.

Explanation

- A **Security Event** is any observable or logged activity, such as:
 - User login attempts
 - File access
 - Process execution
 - Network connections
- A **Security Incident** is a confirmed malicious activity that:
 - Violates security policy
 - Impacts confidentiality, integrity, or availability
- Key differences:
 - Events are **raw signals**
 - Incidents represent **validated risk**
- Most events are **benign or expected**
- SOCs focus on identifying **which events actually matter**

Example:

- 1 failed login → Event
- 50 failed logins from foreign IP → Potential incident

SOC Organization Structure (People)

Goal

Understand how SOC teams are structured in real organizations.

Explanation

- **L1 Analysts:** Monitor dashboards, acknowledge alerts, perform basic validation
- **L2 Analysts:** Perform deeper investigation, log correlation, attack scope analysis
- **L3 Analysts / Threat Hunters:** Proactively hunt for threats, analyze advanced attacks
- **SOC Manager / Lead:** Oversees operations, metrics, escalations, and leadership communication
- **Incident Response Team:** Handles containment, eradication, and recovery
- **Threat Intelligence Team:** Tracks attacker TTPs, enriches IOCs, maintains threat context

SOC models used by organizations:

- Internal SOC
- Managed SOC (MSSP)
- Hybrid SOC

SOC Tools & Technology Stack (Unified View)

Goal

Understand the tools that power a SOC and how they work together.

Explanation

- **SIEM:** Centralizes logs, correlates events, generates alerts
 - Examples: Splunk, Elastic, QRadar, Azure Sentinel
- **EDR/XDR:** Monitors endpoint behavior and detects malicious activity
 - Examples: CrowdStrike, Microsoft Defender, SentinelOne
- **SOAR:** Automates repetitive tasks and orchestrates responses
 - Examples: Cortex XSOAR, Shuffle, Tines
- **Threat Intelligence Platforms:** Enrich IPs, URLs, file hashes
 - Examples: VirusTotal, OTX, ThreatFox
- **Network Sensors:** Provide network-level visibility
 - Firewalls, IDS/IPS, Proxy, DNS filtering

 SOC tools are **integrated**, not used in isolation

SOC Process (How SOC Works End-to-End)

Goal

Understand the standard SOC operating process.

Explanation

- **Log Generation:** Systems generate logs (Windows, Linux, firewall, cloud)
- **Log Forwarding:** Agents send logs to central platforms
- **Ingestion & Normalization:** SIEM parses and structures data
- **Detection & Correlation:** Rules and analytics identify suspicious patterns
- **Validation & Escalation:** Potential threats are validated and escalated
- **Response Coordination:** SOC coordinates actions with IT and security teams
- **Post-Incident Improvement:** Rules, controls, and detections are improved

This process is **continuous and iterative**

Common SOC Alerts & Scenarios

Goal

Understand the types of alerts SOC's commonly see.

Explanation

- Brute-force login attempts
- Malware detection on endpoints
- Suspicious logins (impossible travel, TOR usage)
- Phishing emails reported by users
- Port scanning and reconnaissance
- Privilege escalation attempts

SOC analysts must understand **context**, not just alerts.

SOC KPIs & Business Value

Goal

Understand how SOC performance is measured.

Explanation

- **MTTD:** How quickly threats are detected
- **MTTR:** How quickly action is taken
- **False Positive Rate:** Too many false alerts reduce efficiency
- **Alert Fatigue:** Overloaded analysts miss real threats
- **Business Impact:** Faster detection = lower damage

SOC success is measured by **risk reduction**, not alert volume.